Vinay T

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Summary

Final-year Computer Science & Data Science undergraduate with a strong foundation in C++, Python, SQL, Object-Oriented Programming (OOP), Data Structures & Algorithms, and software development. Experienced in developing AI, computer vision, and cybersecurity applications using Python, OpenCV, YOLO, Scikit-learn, and Scapy. Built real-time systems including a smoke and fire detection platform and an AI-driven network intrusion detection system. Currently building **Sanchari.ai**, an AI-powered travel application using React, Next.js, and intelligent recommendation systems while expanding expertise in LLMs and RAG. Regularly solve LeetCode problems to strengthen problem-solving and algorithmic skills.

Skills

Languages: C++, C, Python, JavaScript
Frontend: React.js, HTML, CSS
Machine Learning: Scikit-learn, YOLOv8, PCA, MLP, Anomaly Detection, Feature Engineering
Cybersecurity: Network Security, Intrusion Detection, Real-Time Packet Analysis, Scapy, Npcap
Data Tools: SQL, Excel, Tableau, Power BI, EDA
CS Fundamentals: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOPS
Backend & Developer Tools: FastAPI, WebSockets, VS Code, Git, GitHub

Projects

SafeML – AI-Driven Real-Time Intrusion Detection System | *Python, Scapy, Npcap, FastAPI, React* **Oct 2025 – Present**

- Developed a real-time network intrusion detection system that captures live packets using **Scapy and Npcap**, converts traffic into network flows, and extracts **78 flow-based features** for analysis.
- Built an ML-based detection pipeline using **MLP, PCA, and feature scaling** to classify network traffic into Benign, PortScan, DDoS, and Bot categories.
- Implemented **anomaly and statistical drift detection** to identify significant deviations in network behavior and flag potential previously unseen or zero-day threats.
- Integrated a **FastAPI + WebSocket backend** with a React dashboard for real-time traffic monitoring, threat scoring, confidence analysis, and security alerts.

Smoke and Fire Detection System using YOLOv8 | *Python, YOLOv8, OpenCV, Google Colab* **Jan 2026 – Apr 2026**

- Built an AI-based real-time smoke and fire detection system using a custom-trained YOLOv8 model for early hazard detection.
- Trained and optimized the model on a large annotated smoke and fire dataset to improve detection accuracy and inference performance.
- Integrated OpenCV for live webcam and video inference with real-time bounding box visualization and confidence scoring.
- Implemented an automatic alarm system to provide instant alerts upon detecting smoke or fire, enhancing safety and emergency response.

Real-Time Air Quality Index Predictor | *Python, Machine Learning, APIs* **Feb 2026 – Mar 2026**

- Built a Python-based ML system that ingests real-time environmental data including PM2.5, PM10, NO_2, and CO through public APIs to predict AQI.
- Performed end-to-end EDA and feature engineering on multi-source environmental datasets to identify key pollutant contributors.
- Implemented Random Forest classification to categorize air quality as Good, Moderate, or Hazardous with Matplotlib-based visualizations.

Awards & Certifications

- **NPTEL – Cloud Computing (2025):** Successfully completed the NPTEL Cloud Computing course with an **Elite** certification, securing a score of **62/100**.
- **Coursera – Data Visualization (2025):** Completed coursework on chart design and data storytelling.
- **Coursera – HTML and CSS in Depth (2025):** Successfully completed the Meta HTML and CSS in Depth course, gaining hands-on experience in responsive web design and modern front-end development.
- **LeetCode:** Solved 100+ problems using C++,MySQL strengthening algorithmic problem-solving and data structures skills.

Education

BE - CSE (Data Science) **Graduating: 2027**

RNS Institute of Technology *CGPA: 8.55 / 10*